

MINSK NATL, Belarus

WMO: 261148

Lat: 53.882N

Lon: 28.031E

Elev: 204

StdP: 98.90

Time Zone: 3.00 (E03)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.1	-17.8	-23.2	0.5	-21.0	-20.0	0.7	-17.1	8.9	-6.4	8.1	-6.2	1.7	310	0.573

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	29.8	19.7	27.8	19.0	25.9	18.3	21.2	26.7	20.2	25.2	19.3	24.1	3.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.2	14.3	23.3	18.2	13.4	22.1	17.8	13.1	21.6	62.7	26.8	58.9	25.1	55.8	23.6	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.3	6.4	DB	-25.1	32.2	4.0	1.7	-28.0	33.4	-30.3	34.5	-32.5	35.4	-35.4	36.7	(4)
(5)			WB	-25.3	22.8	3.9	1.1	-28.2	23.6	-30.4	24.2	-32.6	24.9	-35.5	25.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.7	-5.5	-4.5	0.1	7.1	12.6	16.1	18.6	17.4	12.2	6.3	1.7	-2.5	(6)	
	DBStd	9.50	6.36	6.07	4.89	4.15	4.33	3.45	3.05	3.36	3.66	4.20	4.27	5.19	(7)	
	HDD10.0	2112	481	407	309	106	23	1	0	0	19	129	250	387	(8)	
	HDD18.3	4351	739	640	567	338	184	84	34	59	186	374	500	645	(9)	
	CDD10.0	903	0	0	1	18	103	184	267	231	85	14	1	0	(10)	
	CDD18.3	99	0	0	0	0	6	18	43	31	2	0	0	0	(11)	
	CDH23.3	1083	0	0	0	5	83	199	438	329	29	0	0	0	(12)	
	CDH26.7	272	0	0	0	1	12	37	118	102	2	0	0	0	(13)	
(14) Wind	WSAvg	2.9	3.0	3.1	3.2	2.9	2.9	2.7	2.5	2.4	2.5	2.7	3.0	3.2	(14)	
(15) Precipitation	PrecAvg	631	41	37	38	39	62	74	91	62	50	53	45	40	(15)	
	PrecMax	863	80	75	64	80	117	157	145	173	121	142	92	71	(16)	
	PrecMin	510	7	17	7	3	16	20	8	6	10	2	7	14	(17)	
	PrecStd	97	17	14	16	18	27	34	39	37	27	34	20	15	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.2	6.0	14.9	23.8	28.2	30.0	31.8	33.1	26.0	19.9	12.2	8.2	(19)	
		MCWB	5.9	5.1	8.8	14.1	18.2	19.7	20.4	19.2	17.8	14.5	10.5	7.6	(20)	
	2%	DB	4.0	4.1	11.2	20.2	25.9	27.8	29.9	29.2	23.9	16.2	10.1	6.2	(21)	
		MCWB	3.7	3.3	6.4	12.3	17.1	18.7	20.0	19.4	16.7	13.0	9.2	5.9	(22)	
	5%	DB	2.9	3.0	8.9	17.8	23.2	25.9	27.9	26.9	21.0	14.1	8.8	5.0	(23)	
		MCWB	2.5	2.5	5.4	10.8	15.6	18.0	19.5	18.7	15.6	12.0	8.1	4.7	(24)	
	10%	DB	1.2	2.0	6.2	15.1	21.1	23.9	25.9	24.8	19.0	12.2	7.1	3.2	(25)	
		MCWB	1.0	1.5	4.2	9.2	14.3	17.2	18.9	18.1	14.4	10.5	6.6	2.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.0	5.0	9.7	15.1	19.9	21.7	22.7	22.3	19.4	15.5	11.1	7.8	(27)	
		MCDB	6.2	5.5	13.8	21.6	25.6	26.9	28.2	28.7	23.9	18.5	11.9	8.2	(28)	
	2%	WB	3.5	3.4	7.4	13.0	18.0	20.1	21.5	20.8	17.4	13.8	9.3	6.0	(29)	
		MCDB	3.7	3.7	10.3	18.3	23.6	25.2	27.0	26.3	21.7	15.8	9.9	6.1	(30)	
	5%	WB	2.2	2.3	5.8	11.6	16.7	19.1	20.4	19.7	16.0	12.3	7.8	4.5	(31)	
		MCDB	2.5	2.7	7.7	16.1	22.0	23.9	25.5	24.6	19.8	14.1	8.3	4.8	(32)	
	10%	WB	1.0	1.3	4.3	10.1	15.3	17.8	19.5	18.6	14.9	11.0	6.4	2.9	(33)	
		MCDB	1.3	1.6	6.0	13.7	19.6	22.1	24.2	23.3	18.2	12.5	6.8	3.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.8	5.4	7.8	11.3	11.8	11.4	11.0	11.2	10.3	7.2	4.2	4.0	(35)	
		MCDBR	3.6	4.8	11.3	15.5	14.9	14.4	14.1	14.5	13.7	10.0	5.9	4.2	(36)	
	5% WB	MCWBR	3.6	4.0	7.2	8.3	7.3	6.8	6.0	6.2	7.5	6.9	5.0	4.0	(37)	
		MCWBR	3.7	4.2	9.6	13.5	13.2	12.6	12.1	12.3	12.2	9.1	5.2	4.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.289	0.323	0.355	0.409	0.400	0.385	0.414	0.417	0.378	0.352	0.331	0.291	(40)	
		taud	2.165	2.119	2.159	2.189	2.288	2.357	2.287	2.276	2.346	2.398	2.281	2.202	(41)	
	Ebn at Noon		650	747	811	815	847	864	828	800	787	725	601	569	(42)	
		Edh at Noon	65	96	115	128	123	116	122	117	96	74	60	52	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.68	1.49	2.82	4.07	5.19	5.62	5.29	4.50	3.03	1.53	0.55	0.40	(44)	
	RadStd		0.14	0.28	0.44	0.49	0.52	0.45	0.52	0.42	0.27	0.30	0.08	0.11	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.81	N/A	N/A	N/A	N/A

Nomenclature: See separate page